# URL-Based phishing attack detection

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# Introduction

- Phishing is a growing threat on the Internet, where attackers deceive users to steal sensitive information. This project aims to develop a machine learning model capable of detecting phishing websites with high accuracy. Several models have been tested to evaluate their performance on our dataset, and their results are compared in this report.

# Definition of Ensemble Learning and its Main Techniques

- **Ensemble Learning** is a Machine Learning approach that combines multiple models to improve prediction performance. Instead of relying on a single model, Ensemble Learning aggregates the predictions of multiple models to increase robustness, reduce overfitting, and improve accuracy.
- The three main Ensemble Learning techniques are:

1. **Bagging (Bootstrap Aggregating):**

- Bagging trains multiple independent models in parallel on different subsamples of data, obtained through bootstrapping (random sampling with replacement). The final prediction is made by averaging (for regressions) or majority voting (for classifications).

🔹 **Example :** Random Forest is a bagging-based algorithm that trains multiple independent decision trees.

🔹 **Advantages :** Reduces variance, improves model stability, and decreases overfitting risk.

🔹 **Disadvantages :** Does not reduce bias and can be computationally expensive if many models are used.

2. **Boosting :**

- Boosting trains models sequentially, with each model trying to correct the errors of previous ones by giving more weight to difficult-to-classify errors. Unlike bagging, the models are not independent.

🔹 **Example :** XGBoost, AdaBoost, and LightGBM are high-performance boosting algorithms.

🔹 **Advantages:** Improves accuracy, reduces bias, and works well with imbalanced datasets.

🔹 **Disadvantages :** Sensitive to overfitting and requires careful hyperparameter tuning.

3. **Stacking (Stacked Generalization):**

- Stacking combines multiple heterogeneous models and uses a meta-model to learn how to optimally combine the base models' predictions. Unlike bagging and boosting, stacking seeks to leverage the strengths of each model by learning how to intelligently combine them.

🔹 **Example:** A combination of Random Forest, XGBoost, and LightGBM with logistic regression as the meta-model.

🔹 **Advantages:** Further improves overall performance by exploiting model complementarity.

🔹 **Disadvantages:** Complex to implement and computationally expensive for training.

# Tested Models and Their Functioning

1. **Random Forest(bagging):**

- Random Forest is an ensemble of decision trees that operate based on bagging principles. Each tree is trained on a subsample of data and makes a prediction. The final decision is made by majority voting among the trees.
- **Why this model?** Random Forest is robust against overfitting and provides stable results, making it useful for phishing URL classification.

2. **XGBoost(boosting):**

- XGBoost is a boosting algorithm based on decision trees. It builds trees sequentially, correcting errors from previous trees.
- **Why this model?** XGBoost is known for its high performance and ability to handle imbalanced datasets.

**3.LightGBM(boosting):**

- LightGBM is an optimized variant of gradient boosting models, designed to be faster and more memory-efficient than XGBoost.
- **Why this model?** LightGBM is particularly effective for large datasets and provides similar performance to XGBoost with reduced computational cost.

**4.AdaBoost(boosting):**

- AdaBoost is a boosting algorithm that combines multiple weak classifiers (such as simple decision trees) to create a stronger model.
- **Why this model?** AdaBoost effectively improves overall accuracy while being lighter than other boosting models.

**5. Random Forest + XGBoost + LightGBM + AdaBoost(stacking):**

- This approach combines multiple models by merging their predictions to achieve better robustness.
- **Why this model?** Using multiple complementary models improves accuracy and reduces errors specific to each algorithm.

**6.MLP (Bagging):**

- A Multi-Layer Perceptron (MLP) is an artificial neural network that learns by adjusting connection weights between neurons. When used with bagging, multiple MLPs are trained on different subsets of data.
- **Why this model?** MLP is a powerful model that can learn complex relationships in data, and bagging helps stabilize its predictions.

**7.MLP + CNN + LSTM + XGBoost + Random Forest(stacking):**

- This hybrid model combines an MLP, a Convolutional Neural Network (CNN), a Long Short-Term Memory (LSTM) network, and ensemble tree models.
- **Why this model?** Leveraging multiple architectures allows us to exploit their strengths: CNN captures patterns in URLs, LSTM handles sequences, and ensemble models ensure prediction stability

# Results Found

| Modèle | Accuracy | Recall | Précision | F1-score |
|---|---|---|---|---|
| Random Forest | 0.96 | 0.96 | 0.96 | 0.96 |
| XGBoost | 0.97 | 0.97 | 0.97 | 0.97 |
| LightGBM | 0.96 | 0.96 | 0.96 | 0.96 |
| AdaBoost | 0.93 | 0.93 | 0.93 | 0.93 |
| Random Forest + XGBoost + LightGBM + AdaBoost | 0.97 | 0.97 | 0.97 | 0.97 |
| MLP (Bagging) | 0.97 | 0.97 | 0.97 | 0.97 |
| MLP + CNN + LSTM + XGBoost + Random Forest | 0.97 | 0.97 | 0.97 | 0.97 |

## Conclusion

The results show that ensemble models (XGBoost, LightGBM, and combinations) achieve the best performance. Additionally, advanced neural network models (MLP, CNN, LSTM) also yield excellent results.